



A Polyhydron Group Company

DOUBLE THROTTLE CUM CHECK VALVE Model : 2TCL10

Ref. No. D 04546 B
Release 12 / 2025

ENGINEERING - 1 of 2

Description

Double throttle check valves model **2TCL10** are meant for controlling flow in one direction in circuit by simple throttling of flow. Reverse flow is free and is independent of throttle condition in opposite direction in same path. The valve is available with throttle check valve facility on either A port, B port or A and B ports.

These are non-pressure compensated flow control valves and therefore offer constant flow for a given setting, only if the pressure drop across the throttled passage is constant.

The valve can be converted from meter-in condition to meter-out condition by simply rotating the body along its longitudinal axis by 180 degree, while being installed.

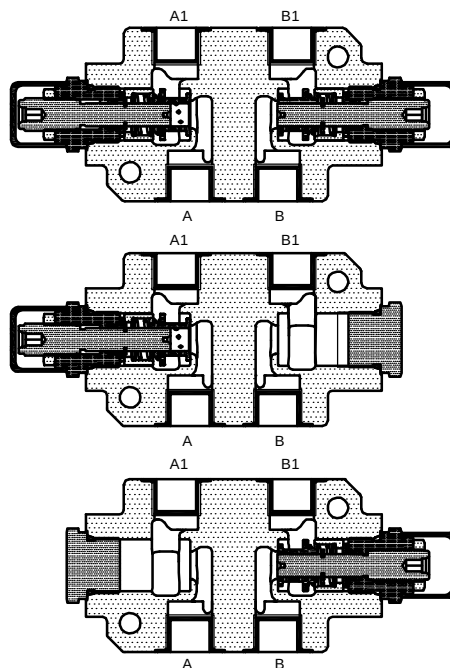
Rotation of the throttling screw in clockwise direction, increase the pressure drop in the path.

For locking the setting, a check nut is provided.

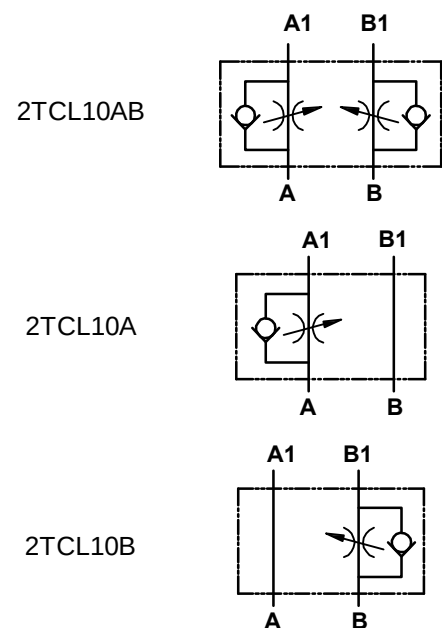
For protection of the setting, a protective cap is provided.



Section



Hydraulic symbol



Technical specifications

Construction	:	Threaded, spool type, Non-pressure compensated.
Mounting	:	Factory standard.
Mounting Position	:	Optional.
Direction of flow	:	Refer Hydraulic symbols.
Flow handling capacity	:	Refer graphs.
Operating Pressure	:	315 bar.
Viscosity range	:	10 cSt to 380 cSt.
Fluid temperature range	:	-20°C to +80°C.
Fluid cleanliness required	:	ISO 4406 20/18/15 or better.
Mass approx.	:	Model 2TCL10AB 2TCL10A/B
		in Kg. 2.1 2

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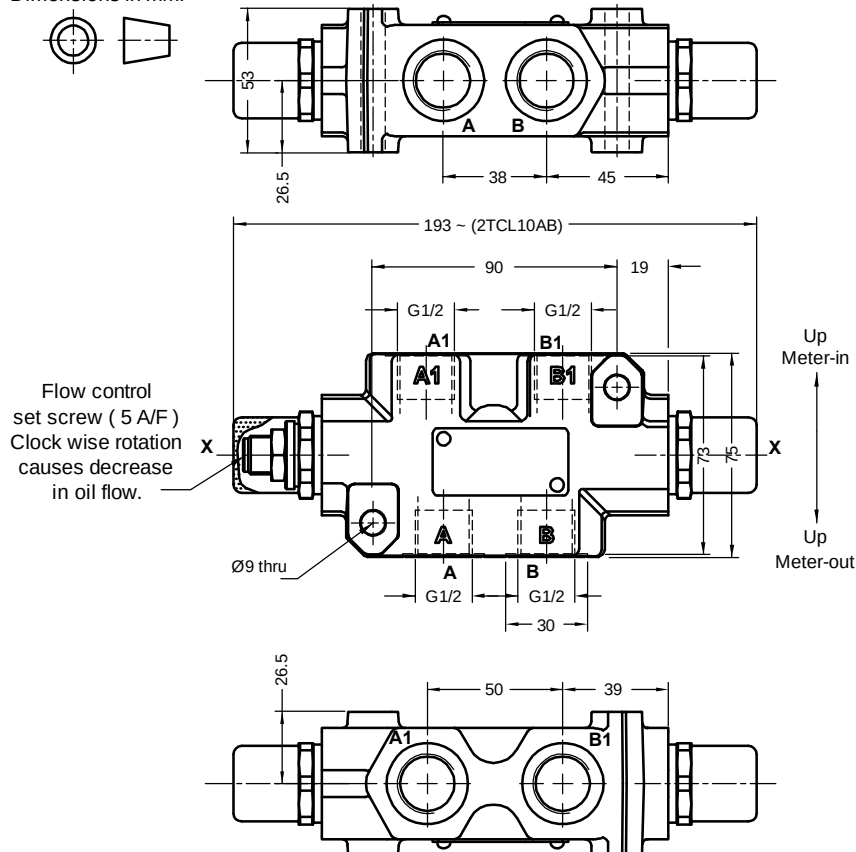
78-80, Machhe Industrial Estate,
Machhe, Belgaum - 590 014. INDIA.

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E-mail : sales@polyhydron.com
Website : www.polyhydron.com



Unit dimensions

Dimensions in mm.

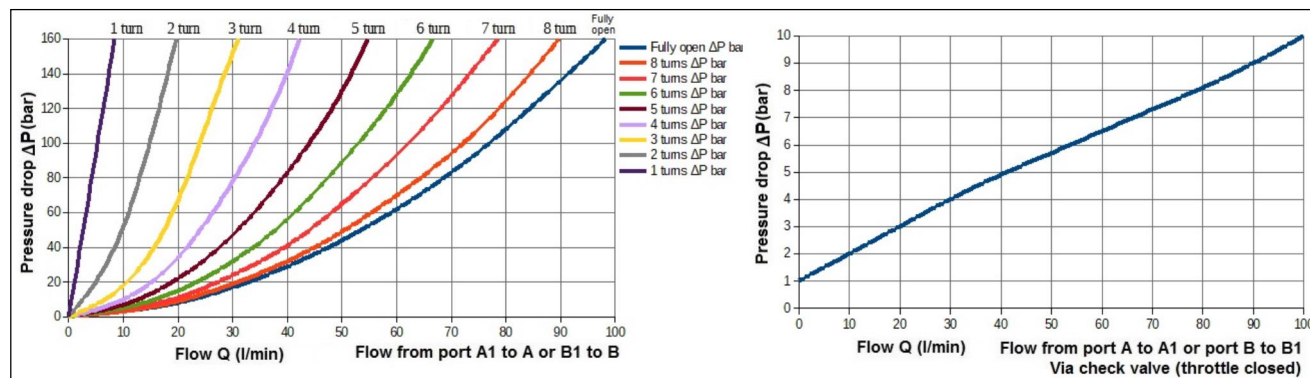


Note :

Rotate the valve around horizontal axis 'X' - 'X' by 180° to convert from METER-IN control to METER-OUT control.

Valve fixing S.H.C Screws are not in scope of supply.
Tightening torque for M8x1.25 S.H.C Screws is 39 Nm.

Performance graph



Ordering code

2TCL 10 AB 01

Double Throttle cum check valve

Size 10

Throttle / check facility	
On A & B ports	AB
On 'A' port only	A
On 'B' port only	B

Design code subjected to Change .
installation dimensions remain same for design code 01 thru 09.

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For critical applications, please ask for certified installation drawing.

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Subject to revision



A Polyhydron Group Company

THROTTLE CUM CHECK VALVE MODULAR CONSTRUCTION

Model : TCM10

Ref. No. D 04546 A

Release 12 / 2025

ENGINEERING - 1 of 2

Description

Throttle check valves model **TCM 10** are meant for controlling flow in one direction in circuit by simple throttling of flow. Reverse flow is free and is independent of throttle condition in opposite direction in same path. The valve is available with throttle check valve facility on either A port, B port or A and B ports.

These are non-pressure compensated flow control valves and therefore offer constant flow for a given setting, only if the pressure drop across the throttled passage is constant.

The valve can be converted from meter-in condition to meter-out condition by simply rotating the body along its longitudinal axis by 180 degree, while being installed.

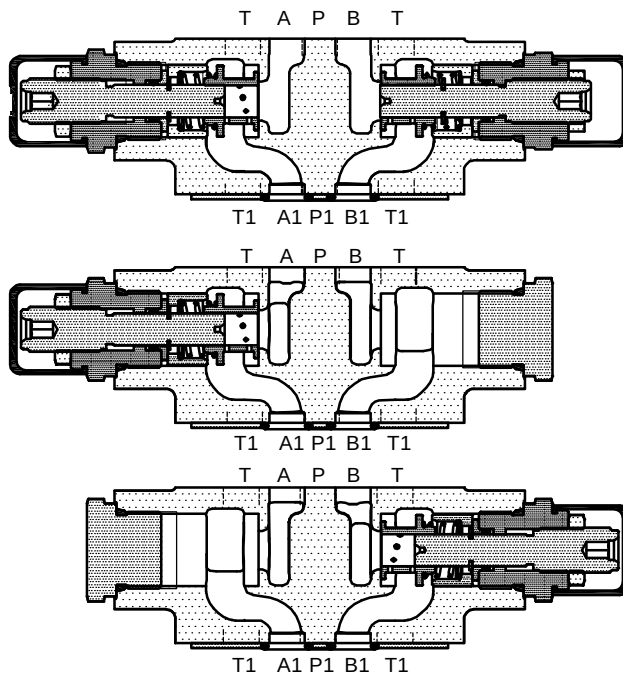
Rotation of the throttling screw in clockwise direction, increase the pressure drop in the path.

For locking the setting, a check nut is provided.

For protection of the setting, a protective cap is provided.

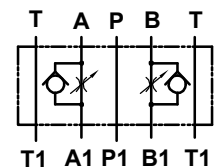


Section

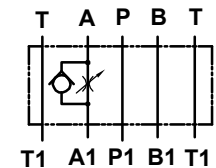


Hydraulic symbol

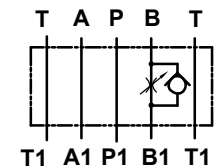
TCM 10 AB



TCM 10 A



TCM 10 B



Technical specifications

Construction	:	Modular, spool type, Non-pressure compensated.
Mounting	:	Modular type, Conforming to 4401 - 05 - 04 - 0 - 94.
Mounting Position	:	Optional.
Direction of flow	:	For ports P and T free flow in either direction. For ports A and B refer Hydraulic symbols.
Flow handling capacity	:	Refer graphs.
Operating Pressure	:	315 bar.
Viscosity range	:	10 cSt to 380 cSt.
Fluid temperature range	:	-20°C to +80°C.
Fluid cleanliness required	:	ISO 4406 20/18/15 or better.
Mass approx.	:	Model TCM10AB TCM10A/B
		in Kg. 2.1 2

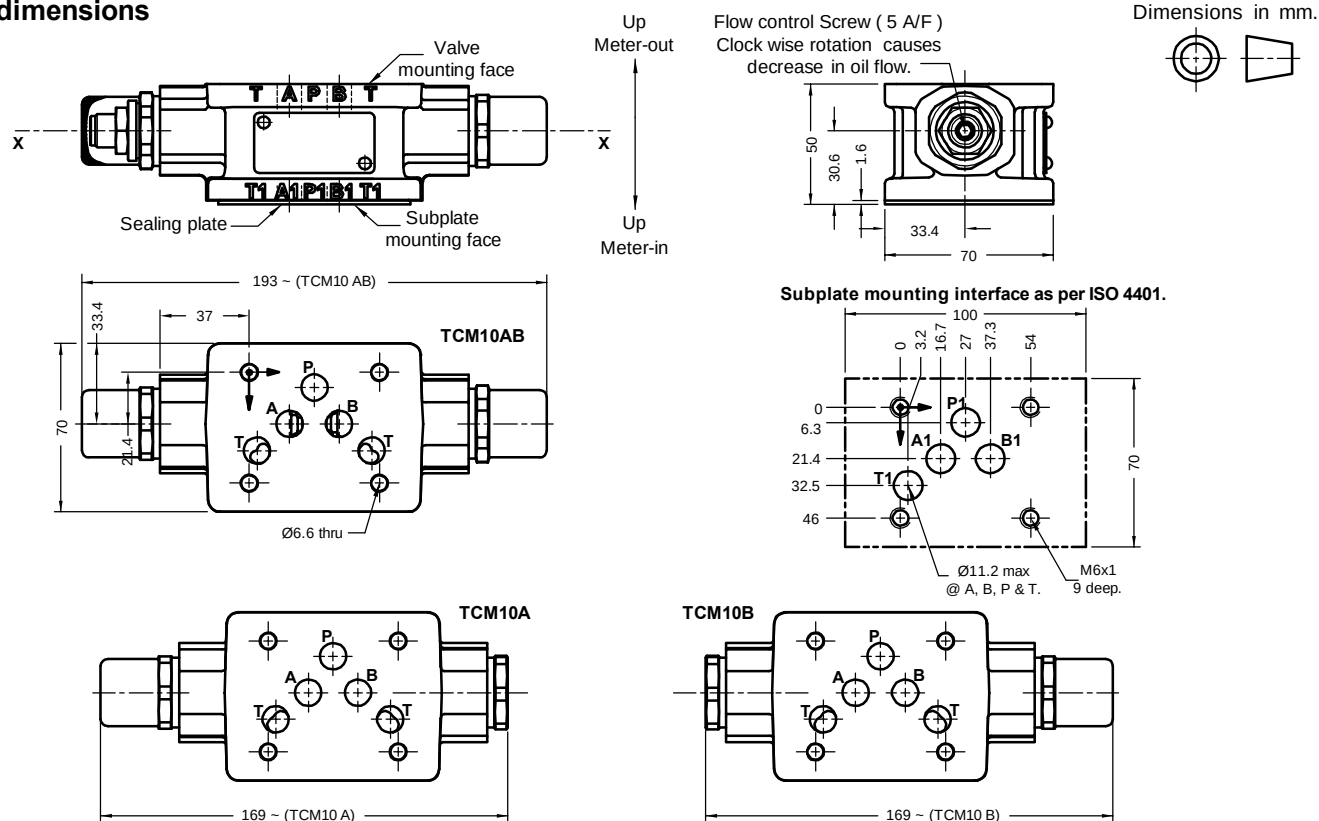
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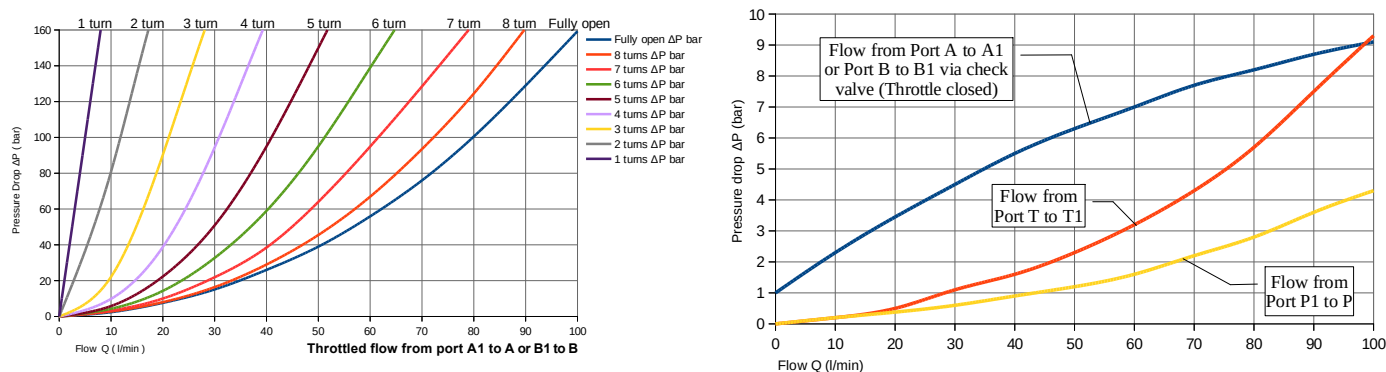


Unit dimensions

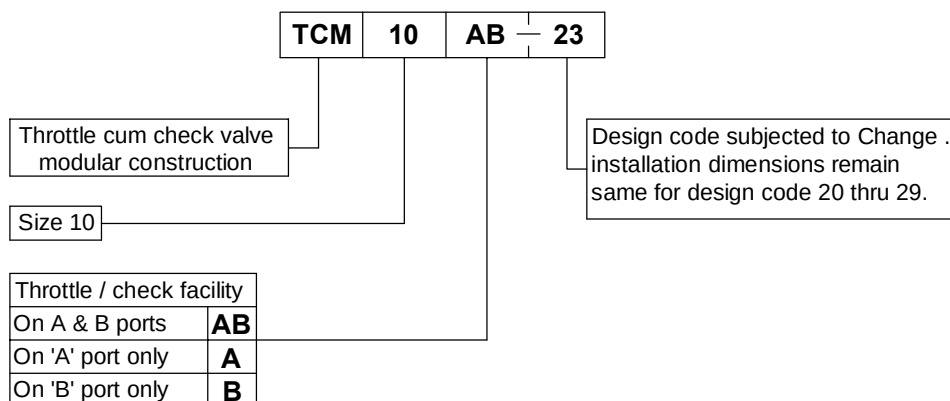


Note : Rotate the valve around horizontal axis 'X' - 'X' by 180° to convert from METER-IN control to METER-OUT control.
Valve fixing S.H.C Screws are not in scope of supply. Tightening torque for S.H.C Screws is 16 Nm.

Performance graph



Ordering code



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A Polyhydon Group Company

THROTTLE / CHECK VALVE MODULAR CONSTRUCTION

Model : TCM06 *

Ref. No. D 05973

Release: 12 / 2025

ENGINEERING - 1 of 2

Description

Throttle check valves model **TCM 06** are meant for controlling flow in one direction in circuit by simple throttling of flow. Reverse flow is free and is independent of throttle condition in opposite direction in same path. the valve is available with throttle check valve facility on either A port, B port or A and B ports.

These are non-pressure compensated flow control valves and therefore offer constant flow for a given setting, only if the pressure drop across the throttled passage is constant.

The valve can be converted from meter-in condition to meter-out condition by simply rotating the body along its longitudinal axis by 180 degree, while being installed.

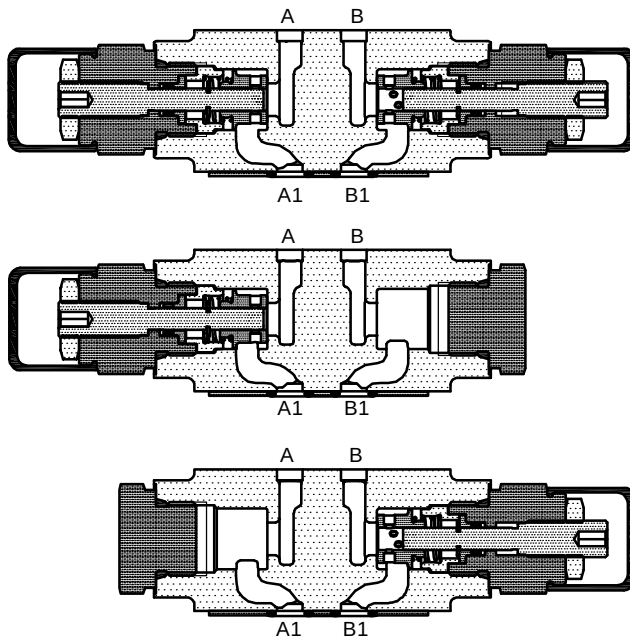
Rotation of the throttling screw in clockwise direction, increase the pressure drop in the path.

For locking the setting, a check nut is provided.

For protection of the setting, a protective cap is provided.

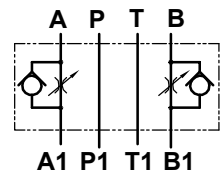


Section

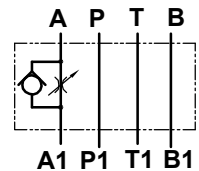


Hydraulic symbol

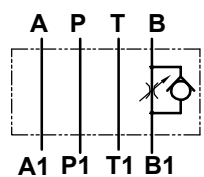
TCM 06 AB



TCM 06 A



TCM 06 B



Technical specifications

Construction	:	Modular, spool type, Non-pressure compensated.
Mounting	:	Modular type, Conforming to 4401-03-02-0-94, IS 10187, DIN24340.
Mounting Position	:	Optional.
Direction of flow	:	For ports P and T free flow in either direction. For ports A and B refer Hydraulic symbols.
Flow handling capacity	:	Refer graphs.
Operating Pressure	:	315 bar.
Viscosity range	:	10 cSt to 380 cSt.
Fluid temperature range	:	-20°C to +80°C.
Fluid cleanliness required	:	ISO 4406 20/18/15 or better.
Mass approx.	:	1.1 Kg.

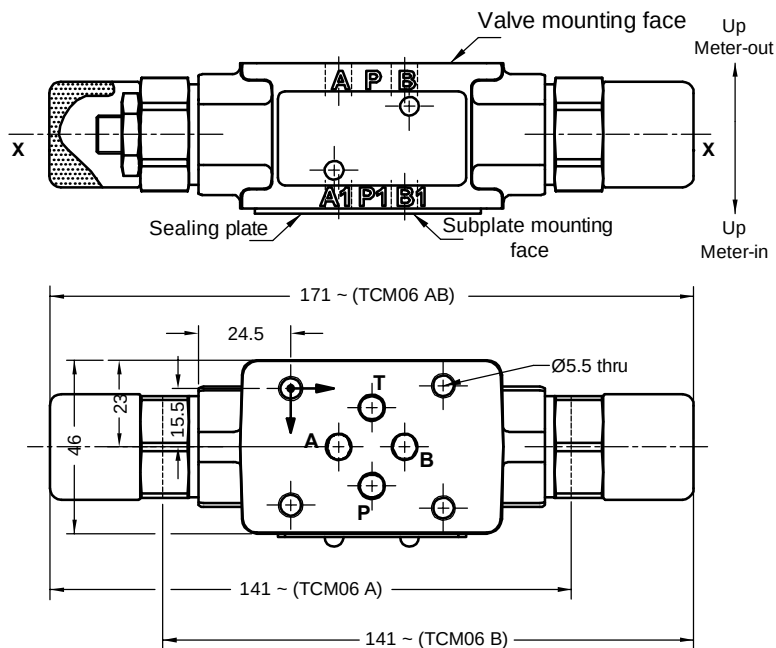
polyhydon pvt. ltd.

78-80, Machhe Industrial Estate,
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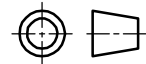
Phone : +91-(0)831- 2411001
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Website : www.polyhydon.com



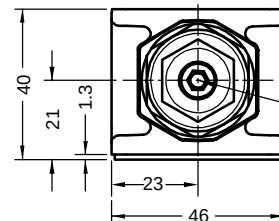
Unit dimensions



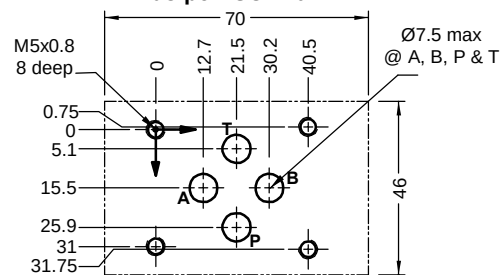
Dimensions in mm.



Flow control
set screw (4 A/F)
Clock wise rotation
causes decrease
in oil flow.



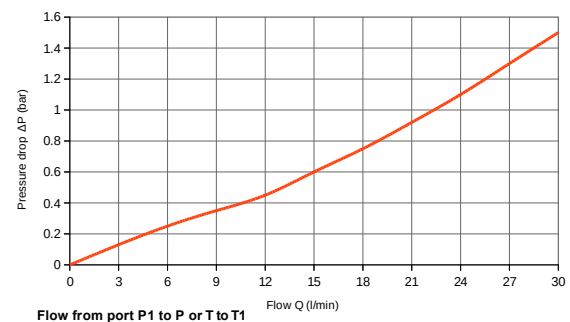
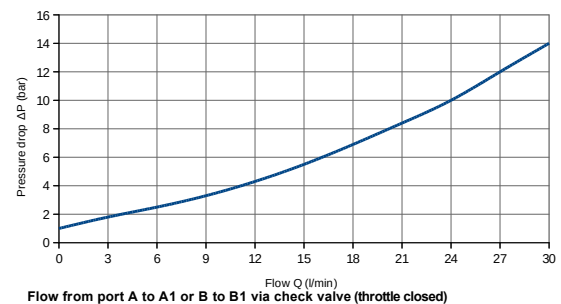
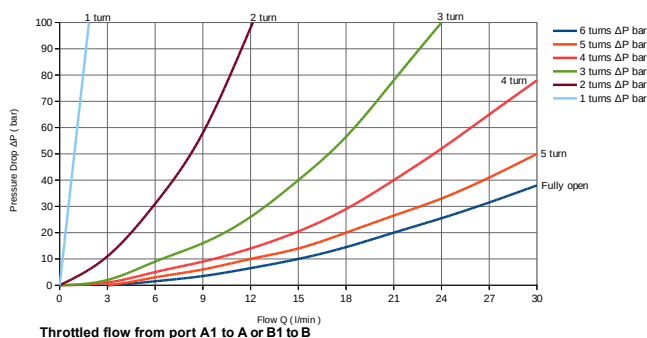
Subplate mounting interface as per ISO 4401.



Note : Rotate the valve around horizontal axis 'X' - 'X' by 180° to convert from METER-IN control to METER-OUT control.
Valve fixing S.H.C Screws are not in scope of supply. Tightening torque for S.H.C Screws is 16 Nm.

Performance graph

Performance test Oil used VG-46 at 30°C.



Ordering code

TCM 06 AB 23

Throttle / check valve
modular construction

Size 06

Throttle / check facility	
On A & B ports	AB
On 'A' port only	A
On 'B' port only	B

Design code subjected to Change .
installation dimensions remain
same for design code 20 thru 29.

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Subject to revision



A Polyhydron Group Company

FLOW DIVIDER Model : FDTA**

Ref. No. D 06727
Release 03 / 2024

ENGINEERING - 1 of 3

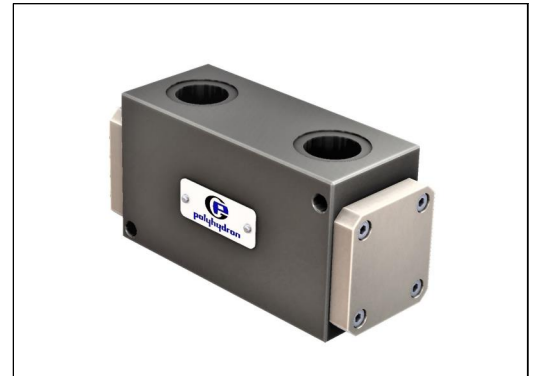
Description

The Flow Divider, model FDTA** are meant for equally dividing the flow supplied to the inlet port P, between outlet port P1 and P2.

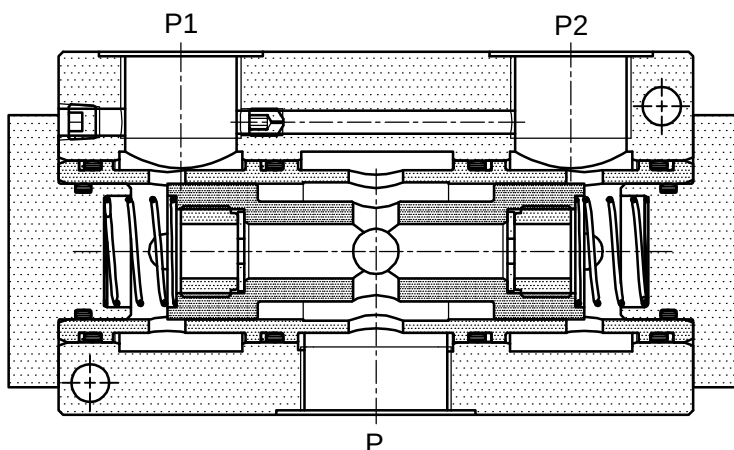
The maximum variation between, dividing the flow can be $\pm 5\%$.

The valve divides the flow irrespective of pressure variation at its outlet ports.

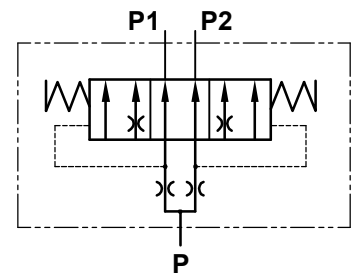
***This is not a flow divider cum combiner valve**, hence, for combining the flow in reverse direction, 'Check Valve' need to be incorporated to bypass the valve.



Section



Hydraulic symbol



Technical specifications

Construction	:	Pressure compensated spool type.
Mounting type	:	Threaded body
Mounting Position	:	Horizontal.
Dividing variation	:	$\pm 5\%$
Working pressure	:	350 bar.
Maximum Flow	:	Refer Ordering code
Hydraulic medium	:	Mineral oil.
Temperature range	:	-20°C to + 80°C.
Viscosity range	:	10 cSt to 380 cSt.
Fluid cleanliness required	:	ISO 4406 20/18/15 or better.
Mass (approx)	:	

Model	NG 10	NG 15
Weight	2.5 kg.	5 kg.

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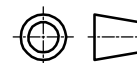
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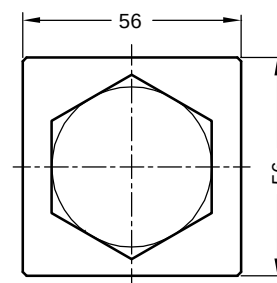
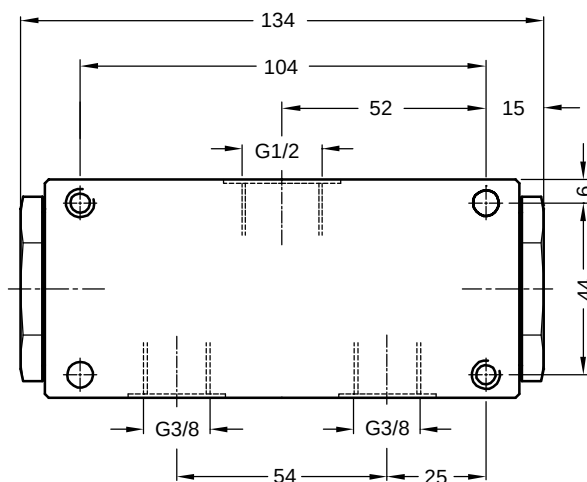


Unit dimensions

Dimensions in mm.

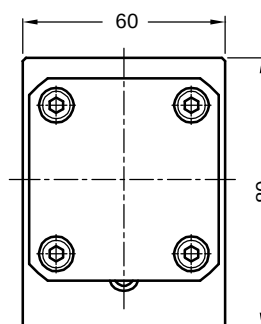
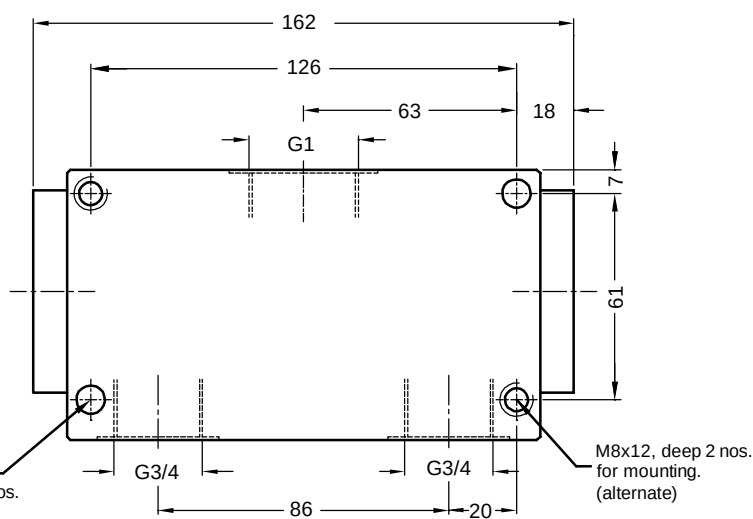


Model : NG 10
FDTAA*
FDTAB*
FDTAC*

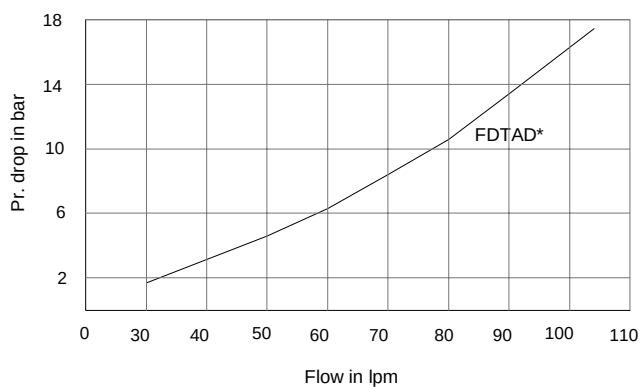
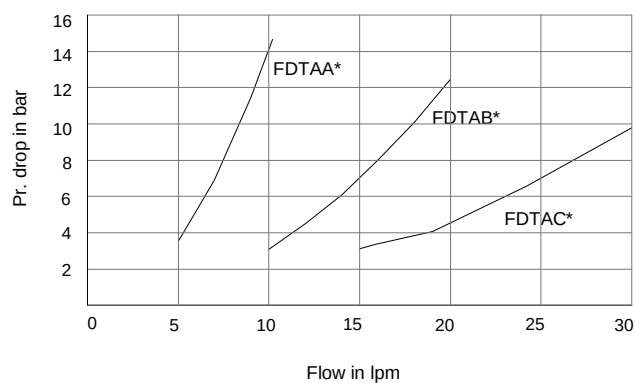


Model : NG 15
FDTAD*

Hole for M8x70L
S.H.C. Screw, 2 nos.
for mounting.

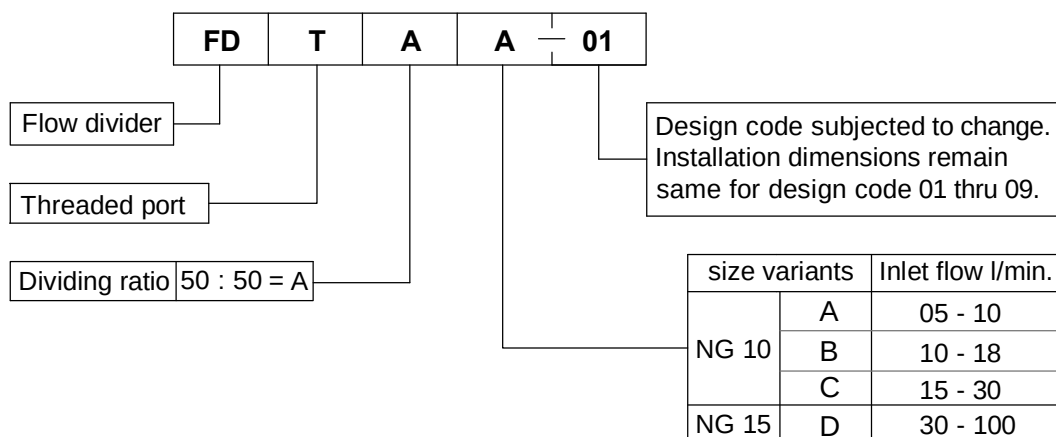


Performance Curves for FDTA**
(Pressure drop related to flow, with 46 cSt at 40°C)





Ordering code





A Polyhydron Group Company

PRESSURE COMPENSATED FLOW CONTROL VALVE Model : 2PF*10****

Ref. No. D 05083

Release 11 / 2025

ENGINEERING - 1 of 3

Description

Pressure compensated flow control in the direction of flow from port A to B.

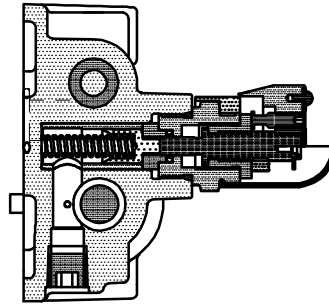
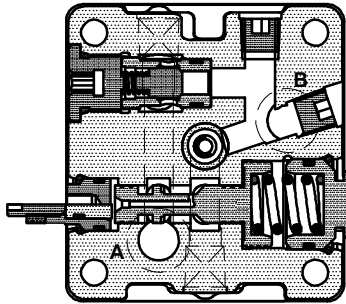
Check valve option for reverse free flow.

Seven rotations of hand knob over the controlling range to facilitate setting with ease.

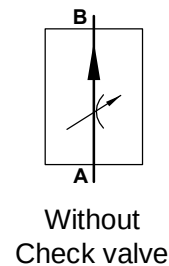
Stroke limiter option available for surge free start at any flow setting.



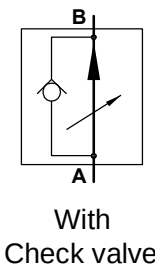
Section



Hydraulic symbol



Without
Check valve



With
Check valve

Technical specifications

Construction	:	Differential piston with sharp edge orifice.
Interface	:	ISO 6263-06-05.
Flow direction	:	Controlled flow from Port A to Port B with optional reverse free check valve.
Flow controlling capacity	:	Minimum 0.25 to 4 l/min, 0.5 to 8 l/min, 1 to 16 l/min or 2 to 32 l/min 3 to 40 l/min.
Flow controlling accuracy	:	± 3%. (At an average temperature of 40°C and Oil viscosity grade 46.)
Operating pressures	:	315 bar.
Minimum pressure differential required	:	7 bar from Port A to Port B.
Hydraulic medium	:	Mineral oil.
Temperature range	:	-20°C to + 80°C.
Viscosity range	:	10 cSt to 380 cSt.
Fluid cleanliness required	:	ISO 4406 20/18/15 or better.
Mass (approx)	:	2.4 Kg.

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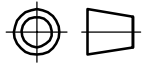
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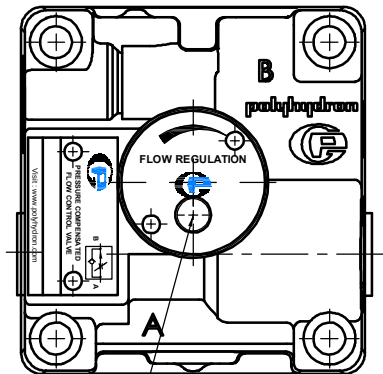
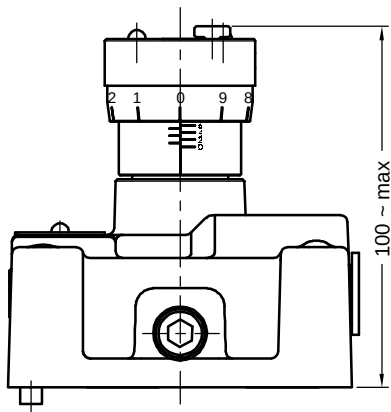


Unit dimensions

Dimensions in mm.

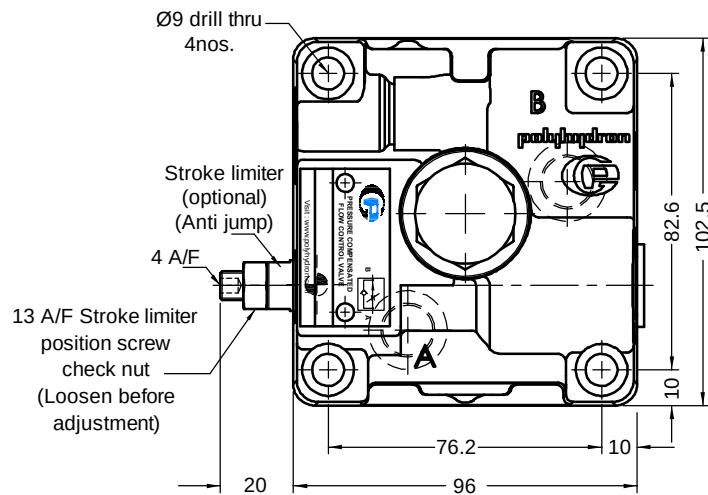
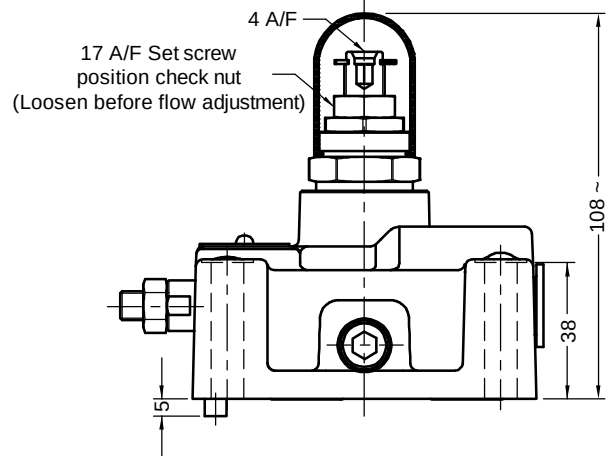


Hand knob type Model : 2PFH10****-14

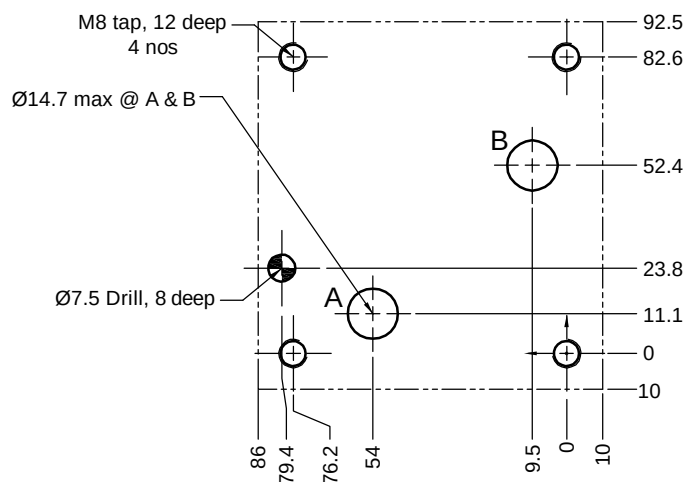


Hand Knob
position locking screw
(Loosen before flow adjustment)

Set screw type Model : 2PFS10****-14

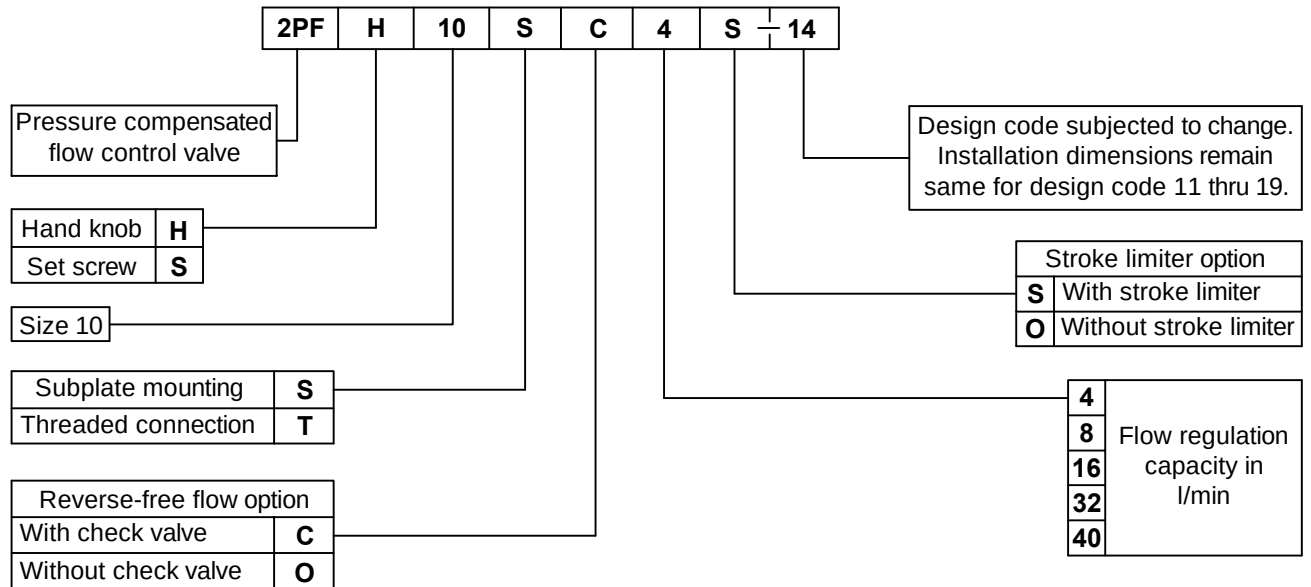


Valve mounting interface

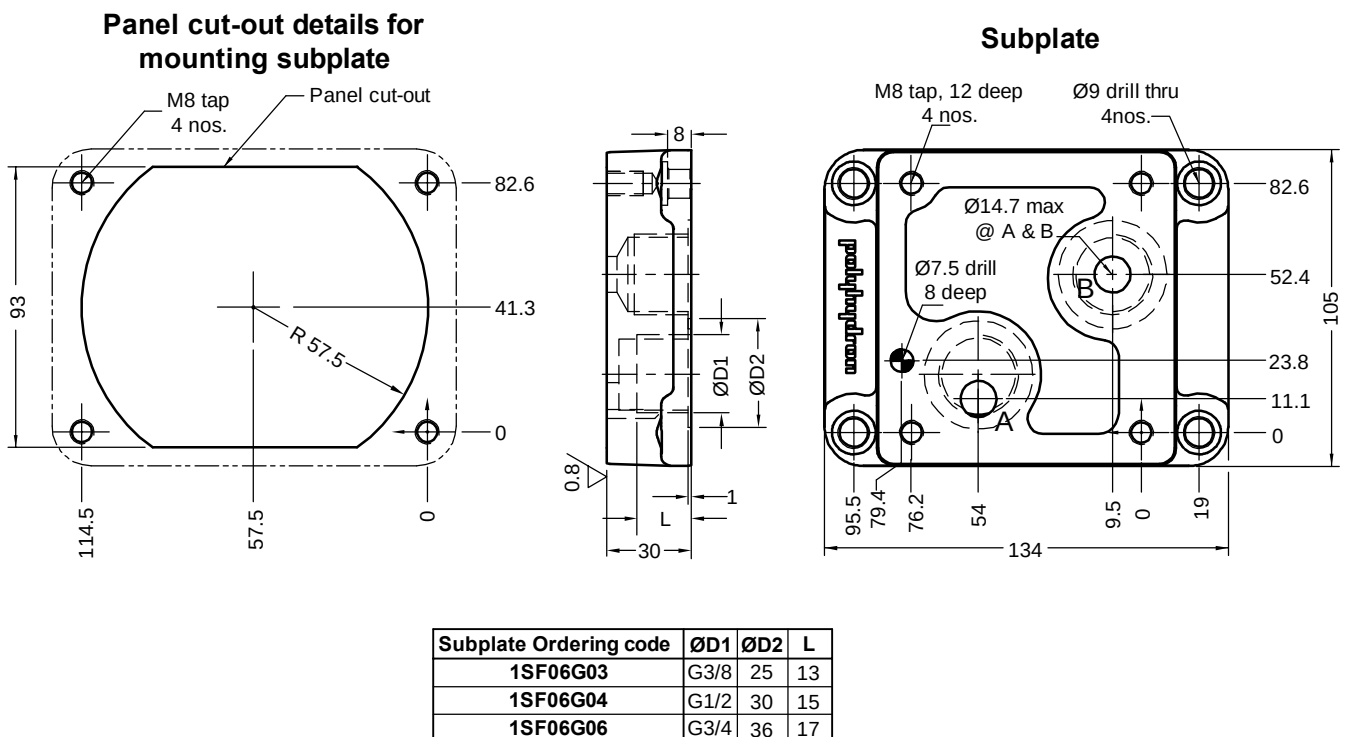




Ordering code



Note : Valve Subplate to be ordered seperately.



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